

# Technical Whitepaper: The HexaHelix B-N-C Architecture

Resonant Magnetohydrodynamics and Direct Energy Conversion in Compact High-Field Stellarators

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## Abstract

The HexaHelix architecture proposes solutions to two major challenges in magnetic confinement fusion: the inefficiency of conventional thermal energy conversion and the manufacturing complexity associated with non-planar high-temperature superconductor (HTS) coils. By implementing a six-fold interlaced helical geometry, optimized through AI-augmented simulations, and employing a Boron-Nitrogen-Carbon (B-N-C) metamaterial, the system can stabilize magnetic fields surpassing 30 tesla within compact volumes. A new Direct Energy Conversion (DEC) module enables resonant inductive extraction of kinetic energy from charged particles, replacing the traditional Carnot-limited thermal cycle. Recent quantum studies on engineered reservoirs and non-local energy transfer support the feasibility of low-loss transmission in the DEC approach. Early simulations suggest efficiencies up to 70%, along with notable reductions in system size and cost, making modular and commercially feasible fusion reactors possible.

## 1. Introduction

Magnetic confinement fusion has advanced considerably, with devices like Wendelstein 7-X achieving notable quasi-isodynamic (QI) performance. Nevertheless, commercial scalability is limited by significant thermal losses and the manufacturing difficulties posed by non-planar HTS coils subjected to extreme mechanical stress. The HexaHelix method, developed through the human-AI collaborative Twin-Code methodology, introduces a topology that integrates geometric resilience, innovative materials, and direct energy extraction. This architecture is designed to transition stellarators from research instruments to efficient, modular solid-state energy generators.

## 2. System Architecture: 6-Fold Interlaced Helix

The main magnet configuration consists of six identical helical modules, interlaced and arranged in a hexagonal symmetry:

- Golden Ratio Twist ( $\varphi \approx 0.618$ ): Enhances magnetic shear and magnetohydrodynamic (MHD) self-stabilization, while minimizing Lorentz force shear stresses.

- **Parallel Manufacturing:** Utilizing six identical modules enables parallel production lines, greatly reducing lead times and complexity compared to unique, large-scale assemblies.
- **Redundancy by Design:** The interlaced arrangement provides fault tolerance, with simulations demonstrating stable confinement even if some segments fail, thanks to compensatory field superposition.
- **Scaling Law:** At magnetic field strengths exceeding 30 T, the required plasma volume for achieving  $Q > 1$  decreases significantly, allowing for a more compact reactor footprint.

### 3. Materials Science: B-N-C Strain Engineering & ISD-Platform

The HexaHelix coils incorporate a B-N-C metamaterial architecture, produced through scalable Chemical Vapor Deposition (CVD):

- **ISD-MgO Advantage (No-Plasma):** The design is based on an ISD-MgO (Inclined Substrate Deposition) template, which enables biaxial texturing without destructive ion bombardment typical in IBAD processes, thereby maintaining the atomic purity of the sensitive sp<sup>2</sup>-hybrid lattice.
- **Internal Pressure Hack:** The synergistic co-doping of Carbon lattices with Boron-11 and Nitrogen generates localized electric fields that simulate multi-gigapascal external pressure. This atomic-scale strain engineering increases the critical temperature and magnetic field stability, eliminating the need for extensive steel containment.
- **Resilient Supply Chain:** Unlike conventional REBCO tapes, the B-N-C approach uses abundant, low-cost industrial precursors such as hydrocarbons, enriched boric acid, and melamine, resulting in reduced material capital expenditures and a more resilient supply chain.
- **Neutron Resilience:** The exclusive use of Boron-11 ( $\sigma \approx 0.005$  barns) minimizes activation and radiation damage compared to natural boron.
- **Thermal Management:** The coils feature an intrinsic 3D-porous architecture (~100 nm pores), which allows efficient coolant flow and achieves ten times greater heat rejection compared to solid coils.

### 4. Energy Extraction: Inductive Direct Energy Conversion (DEC)

Traditional fusion reactors depend on neutron capture and thermal cycles that are limited by Carnot efficiency (~40%). The HexaHelix design introduces resonant inductive coupling for direct energy conversion, acting as an electromagnetic damper for plasma oscillations.

- **Dual Circuit Design:** Two separate coil subsystems, both leveraging B-N-C material but with distinct structures, are used:
- **Magnet Circuit (DC – Steady-State Confinement):** Features a dense, compact B-N-C layer on an ISD-MgO substrate to generate a stable magnetic field (>30 T) for plasma confinement. This layer is solid-state, with no pores, to maximize density and stability, supported by the internal pressure hack for simulating gigapascal pressures. DC operation demands stability with constant current.

- **Generator Circuit (AC – Energy Extraction):** Utilizes a nano-porous B-N-C architecture (~100 nm pores) as a liner facing the plasma, also on the ISD-MgO substrate. This layer acts as a resonant receiver for DEC, using plasma oscillations for inductive energy conversion. The pores optimize flux penetration and coupling. AC operation requires resonance and damping to extract energy and stabilize plasma.

**Integration & Benefits:** Both circuits share the ISD-MgO substrate at room temperature, minimizing mismatches such as thermal stresses and cracks. The generator is positioned as the inner layer (plasma-facing), while the magnet forms the outer stable core, preventing interference between DC and AC operations. The generator damps plasma oscillations, reducing quench risks for the magnet. Projected efficiency exceeds 70%, with compact scaling (e.g., Power-Box dimensions of 2x2x4 m).

**Resonance Condition:** The natural frequency of the B-N-C lattice is tuned to match plasma oscillation frequency, maximizing mutual inductance through the porous structure.

**Direct Efficiency:** Theoretical models indicate up to 70% efficiency by directly converting kinetic energy to induced voltage, eliminating the need for steam turbines.

## 5. Economic Scaling: The 150 MW Pilot Module

The 150 MW industrial pilot module is designed as the smallest commercially viable unit, optimized for factory production and standard logistics:

- **Factory-Built & Modular:** Modules (~5m x 6m) are suitable for parallel manufacturing and on-site torus integration, simplifying transport logistics.
- **CAPEX Reduction:** By removing the need for complex steam infrastructure and using standard industrial CVD equipment, significant capital expense savings are achieved.
- **LCOE Target:** Miniaturization and direct conversion enable an optimized Levelized Cost of Electricity (LCOE) for decentralized, high-availability energy markets.

## 6. Conclusion

The HexaHelix B-N-C architecture reimagines fusion energy from "brute force" engineering to "smart materials" engineering. By integrating AI-optimized geometry, leveraging the manufacturing advantages of ISD-MgO, and implementing resonant energy extraction, this approach offers exponential potential for modular, economically viable fusion energy.

**Open Repository:** [github.com/Haegar1601/hexa-helix-stellarator](https://github.com/Haegar1601/hexa-helix-stellarator)

**License:** Dual MIT / CERN-OHL-W-2.0

## References (selected)

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This whitepaper is generated through Twin-Code Strategy (Human Intuition + AI Logic). Feedback and collaboration welcome.